

Supplement to
Modeling the Modeler

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July 26, 2026

ABSTRACT

This supplemental material to [Payró and Piermont \(2026\)](#) expands on several aspects of the paper. We rigorously examine the identification of expected utility preferences as introduced in Example 1. We then provide two additional examples illustrating the role of concordance, examine the tension between balanced reductions in uncertainty and posterior concentration, and fill in the numerical details for the identification of discount factors in Section 5.1.

1 EXPECTED UTILITY

This section expands on Example 1. For completeness, we restate the setup and notation.

Consider a modeler who believes that the subject has expected utility preferences and can ask the subject to make only a single choice from one menu. In this case, the set of choice patterns $\mathbb{Z}$ is simply the ℓ -dimensional simplex of all relevant lotteries, denoted by Δ , and the set of experiments is $\mathbb{E}^\Delta = \{(A, \mathcal{P}) \mid A \subseteq \Delta \text{ is finite, } \mathcal{P} \text{ partitions } A\}$.

The set of types is in one-to-one correspondence with the set of normalized expected utility functions over Δ . Thus,

$$\Theta^\Delta = \{\theta \in \mathbb{R}^{\ell+1} \mid \theta_{\ell+1} = 0\}$$

and the choice function for each $\theta \in \Theta$ is determined by utility maximization:

$$c_\theta(A) \in \arg \max_{p \in A} \sum_{i=1}^{\ell+1} \theta_i p_i.$$

Let Ω^Δ be the smallest algebra on Θ^Δ containing $W_{A,B}$ for all finite $A \subseteq \Delta$ and $B \subseteq A$, and μ a probability measure over $(\Theta^\Delta, \Omega^\Delta)$.

We observe that the restriction to offering only a single menu in the previous example is without loss of generality. Although in many environments observing only one-shot behavior is limiting, this is not the case under expected utility. The reason is that for any finite

$A, A' \subseteq \Delta$ and linear $u : \Delta \rightarrow \mathbb{R}$,

$$\left. \begin{array}{l} p \in \arg \max_A u(\cdot) \\ q \in \arg \max_{A'} u(\cdot) \end{array} \right\} \text{ if and only if } \alpha p + (1 - \alpha)q \in \arg \max_{\alpha A + (1 - \alpha)A'} u(\cdot)$$

where $\alpha A + (1 - \alpha)A'$ is the weighted Minkowski sum of A, A' .¹ Hence, observing a single choice from $\alpha A + (1 - \alpha)A'$ is equivalent to observing a choice from A and A' and thus, it is without loss of generality for the theory to only consider offering the subject a single menu to make a choice from.

This specialization of our model is closely related to the setup of random expected utility models à la [Gul and Pesendorfer \(2006\)](#) (henceforth, GP) and indeed, the experimenter's prior μ defines a GP random expected utility model.² GP show that a belief μ can be identified from random choice data exactly when it is *regular*, that is, if for any distinct $p, q \in \Delta$, $\mu(\{\theta \in \Theta^\Delta \mid \sum_i \theta_i p_i = \sum_i \theta_i q_i\}) = 0$.

The rest of the section is divided into two parts. First, we prove Proposition 1 of the main text, establishing that regularity of the modeler's prior ensures concordance. Second, we show how structural invariance can be recast as Translation Invariance.

1.1 CONCORDANCE

Recall that an experimental paradigm $(\mathbb{Z}, \mathbb{T}, \mathbb{E})$ is *concordant* if:

- (C1) For all $(A, \mathcal{P}) \in \mathbb{E}$ and all $P \in \mathcal{P}$, $W_{A,P} \in \Omega$.
- (C2) For any finite Ω -measurable partition $\mathcal{W}$ of Θ , there exists an experiment $(A, \mathcal{P}) \in \mathbb{E}$ such that $\{W_{A,P} \mid P \in \mathcal{P}\} \approx_\mu \mathcal{W}$.

PROPOSITION 1. *If μ is regular, then $(\Delta, (\Theta^\Delta, \Omega^\Delta, \mu), \mathbb{E}^\Delta)$ is concordant.*

Proof of Proposition 1. Throughout the proof, we identify the normalized utility space Θ^Δ with $\mathbb{R}^\ell$ and take all normal cones relative to this identification.

For a set $X \subseteq \mathbb{R}^\ell$, let $\text{conv}(X)$, $\text{int}(X)$, and $\text{cl}(X)$ denote the convex hull, the interior, and the closure of X , respectively. If X is convex, then $\text{ext}(X)$ collects all the extreme points of X and $\text{ri}(X)$ denotes the relative interior of X . When it is not confusing to do so, we will write $\text{ri}(X)$ and $\text{ext}(X)$ to mean $\text{ri}(\text{conv}(X))$ and $\text{ext}(\text{conv}(X))$ for non-convex X .

For convex X , let $F \subset X$ be called a face if $\alpha x + (1 - \alpha)y \in F$ (for $x, y \in X$ and $\alpha \in (0, 1)$) implies $x, y \in F$. Let $\mathbb{F}(X)$ denote the set of all (non-empty) faces of X and $\mathbb{F}^\circ(X) = \{\text{ri}(F) \mid F \in \mathbb{F}(X)\}$. If $X \subseteq \mathbb{R}^\ell$ is a convex set and $\text{ext}(X)$ is finite then X is called a polytope. Let poly denote the set of all polytopes in $\mathbb{R}^\ell$. If $X \in \text{poly}$, then $\mathbb{F}(X)$ is finite.

If $K \subseteq \mathbb{R}^\ell$ and $\lambda K \subseteq K$ for all $\lambda \geq 0$ then K is called a *cone*. We say a cone K is generated by X if $K = \{\lambda x \mid x \in X, \lambda \geq 0\}$. A cone K is *polyhedral* if it is generated by a polytope; let $\mathcal{K}^*$ denote all such cones. Let $\mathcal{K}$ denote the set of pointed polyhedral cones, those cones with $\mathbf{0} \in \text{ext}(K)$. The face of a polyhedral cone is a polyhedral cone.

¹That is, $\alpha A + (1 - \alpha)A' = \{\alpha p + (1 - \alpha)q \mid p \in A, q \in A'\}$ for all $\alpha \in [0, 1]$ and $A, A' \subseteq \Delta$.

² Ω^Δ coincides with the algebra of utilities defined in GP.

For $X \in \text{poly}$, let

$$X^* = \bigcup_{I \subseteq \text{ext}(X)} \sum_{i \in I} \frac{i}{|I|}$$

The set X^* is a set that contains one point in the relative interior of every face of X . Further, given a partition $\mathcal{H} = \{H_1 \dots H_n\}$ of $\mathbb{F}(X)$, let $\mathcal{H}^* = \{H_1^* \dots H_n^*\}$ denote the partition of X^* defined via $H_i^* = X^* \cap (\bigcup_{F \in H_i} \text{ri}(F))$.

For $X \subseteq \mathbb{R}^\ell$ (convex or not) and $x \in X$ let $N(X, x) = \{u \in \mathbb{R}^\ell \mid \langle u, (y - x) \rangle \leq 0, \text{ for all } y \in X\}$ denote the normal cone of X at x . Notice that under the experimental paradigm $(\Delta, (\Theta^\Delta, \Omega^\Delta, \mu), \mathbb{E}^\Delta)$,

$$W_{A,B} = \bigcup_{x \in B} N(A, x)$$

for all finite $A, B \subseteq \Delta$.

For $X \in \text{poly}$, and a face $F \in \mathbb{F}(X)$, let $N(X, F) = \bigcap_{x \in F} N(X, x)$. It follows that $N(X, F) = \{u \in \mathbb{R}^\ell \mid F \subseteq \arg \max_{x \in X} \langle u, x \rangle\} = N(X, x)$ for any $x \in \text{ri}(F)$. Notice therefore that

$$\bigcup_{x \in H_i^*} \text{ri}(N(X^*, x)) = \bigcup_{F \in H_i} \text{ri}(N(X, F)). \quad (1)$$

It is immediate that $N(X, F)$ is closed and in $\mathcal{K}^*$. Let $\mathcal{N}(X) = \{N(X, F) \mid F \in \mathbb{F}(X)\}$ denote the normal fan of X ; that $\mathcal{N}(X)$ is a fan indicates that it is a family of cones with the following two properties:

- (i) Every nonempty face of a cone in $\mathcal{N}(X)$ is also a cone in $\mathcal{N}(X)$.
- (ii) The intersection of any two cones in $\mathcal{N}(X)$ is a face of both.

Furthermore, $\mathcal{N}(X)$ is complete (the union of $\mathcal{N}(X)$ is Θ^Δ). Let

$$\mathcal{N}^\circ(X) = \{\text{ri}(N(X, F)) \mid F \in \mathbb{F}(X)\}.$$

LEMMA 1. *The following are true for all convex X and Y :*

1. $\text{cl}(\text{ri}(X)) = \text{cl}(X)$ (Theorem 6.3 of [Rockafellar \(1970\)](#)).
2. $\mathbb{F}^\circ(X)$ is a partition of X (Theorem 18.2 of [Rockafellar \(1970\)](#)).
3. Let $F \in \mathbb{F}(X)$ and $Y \subseteq X$ be such that $\text{ri}(Y) \cap F \neq \emptyset$, then $Y \subseteq F$. (Theorem 18.1 of [Rockafellar \(1970\)](#)).

LEMMA 2. *For polytopes X and X' , the following are equivalent:*

- (i) $X = \alpha X' + Z$, for some polytope Z and $\alpha > 0$
- (ii) for all $K \in \mathcal{N}(X)$ there is a $K' \in \mathcal{N}(X')$ such that $K \subseteq K'$
- (iii) $\mathcal{N}^\circ(X)$ refines $\mathcal{N}^\circ(X')$.

Proof. (i) $\leftrightarrow$ (ii) Theorem 15.1.2 of [Grünbaum \(2003\)](#)

(ii) $\rightarrow$ (iii) Take some $\text{ri}(K) \in \mathcal{N}^\circ(X)$. Let $K^\dagger = \bigcap \{K' \in \mathcal{N}(X') \mid K \subseteq K'\}$, which is an element of $\mathcal{N}(X')$ by (ii) and the properties of fans. Moreover, by construction

$K \not\subseteq F$ for any $F \in \mathbb{F}(K^\dagger)$ with $F \subsetneq K^\dagger$. Thus, by (the contrapositive of) Lemma 1.3, we have that $\text{ri}(K) \cap F = \emptyset$ for all such F . Then, since $\mathbb{F}^\circ(K^\dagger)$ partitions $K^\dagger$, it follows that $\text{ri}(K) \subseteq \text{ri}(K^\dagger)$.

(iii) $\rightarrow$ (ii) Take some $K \in \mathcal{N}(X)$. Then by (iii) $\text{ri}(K) \subseteq \text{ri}(K')$ for some $K' \in \mathcal{N}(X')$. Since K and K' are closed, we have $K = \text{cl}(K)$ and $K' = \text{cl}(K')$. Thus, $K = \text{cl}(\text{ri}(K)) \subseteq \text{cl}(\text{ri}(K')) = K'$, where both equalities come from Lemma 1.1 and the inclusion relation from the fact that taking closures is subset preserving. ★

(C1) follows from the definition of Ω^Δ and $\mathbb{E}^\Delta$. We will show (C2): that any partition can be captured up to μ -equivalence. Let $\mathcal{W} = \{W_1, \dots, W_n\} \in \text{part}(\Theta^\Delta)$. First, from Gul and Pesendorfer (2006) Proposition 6 (ii), we can write each $W_i \in \Omega$ as the finite union of elements in $\mathcal{K}$: $W_i = \bigcup_{j=1}^{m_i} \text{ri}(K_i^j)$. Moreover, by Gul and Pesendorfer (2006) Proposition 4, each $K_i^j = N(X_i^j, x_i^j)$ for some polytope X_i^j and $x_i^j \in X_i^j$. Thus $\text{ri}(K_i^j) \in \mathcal{N}^\circ(X_i^j)$.

Let $a = m_1 + \dots + m_n$, where m_i is the number of cones used to represent W_i , and consider the polytope $X = \sum_{i=1}^n \sum_{j=1}^{m_i} \frac{1}{a} X_i^j$. By Lemma 2, $\mathcal{N}^\circ(X)$ refines each $\mathcal{N}^\circ(X_i^j)$. Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a partition of $\mathbb{F}(X)$ defined by

$$H_i = \{F \in \mathbb{F}(X) \mid \text{ri}(N(X, F)) \subseteq W_i\} \quad (2)$$

Now take some $i \leq n$ and $\theta \in W_i$. So, there exists some $j \leq m_i$ such that $\theta \in \text{ri}(K_i^j) \in \mathcal{N}^\circ(X_i^j)$. Since $\mathcal{N}^\circ(X)$ is a partition of Θ^Δ , there exists some $F \in \mathbb{F}(X)$ with $\theta \in \text{ri}(N(X, F))$, and furthermore, since this partition refines $\mathcal{N}^\circ(X_i^j)$, $\text{ri}(N(X, F)) \subseteq \text{ri}(K_i^j) \subseteq W_i$. Hence $F \in H_i$ and so $\theta \in \bigcup_{F \in H_i} \text{ri}(N(X, F))$. We have established that $W_i \subseteq \bigcup_{F \in H_i} \text{ri}(N(X, F))$, and since the other inclusion is obvious, it follows that $W_i = \bigcup_{F \in H_i} \text{ri}(N(X, F))$. Now, on the basis of (1), we have

$$W_i = \bigcup_{x \in H_i^*} \text{ri}(N(X^*, x)) \quad (3)$$

Finally, by Lemma 2 of Gul and Pesendorfer (2006), we know that for any regular μ , it must be that $\mu(\text{ri}(N(X^*, H^*))) = \mu(N(X^*, H^*))$. Thus, $\{W_{X^*, H^*}\}_{H^* \in \mathcal{H}^*} \approx_\mu \mathcal{W}$. ■

1.2 TRANSLATION INVARIANCE

In this section we show how to recast structural invariance in a domain-specific manner, illuminating that the concrete notion of invariance that is relevant to the expected utility model is *translation invariance*.

Consider the experiment $(A, \{P_1, \dots, P_n\})$ and some other B ; notice that the linearity of each u requires that $u \in W_{A, P_i}$ if and only if $u \in W_{\alpha A + (1-\alpha)B, \alpha P_i + (1-\alpha)B}$. Indeed, there must be some $p \in P_i$ that maximizes u over A and $q \in B$ that maximizes u over B , so that $\alpha p + (1-\alpha)q \in \alpha P_i + (1-\alpha)B$ maximizes u over the mixture. Hence, translating an experiment by mixing both the decision problem and the observability partition with some common A' does not alter which preference sets can be identified. This particular form of

invariance is captured by the following axiom.³

(B1) *Translation Invariance.* For all $A, B \subseteq \Delta$ we have

$$(A, \{P_1, \dots, P_n\}, c) \succeq (\alpha A + (1 - \alpha)B, \{Q_1, \dots, Q_n\}, c') \iff c' \geq c$$

whenever $Q_i \subseteq (\alpha P_i + (1 - \alpha)B)$ for all $i \leq n$, with indifference when $c' = c$.

Recall that Structural Invariance also implies that the value of identification should not depend on zero-probability perturbations. The following axiom reflects this implication.

(B2) *Belief Consistency.* Fix $A \subseteq \Delta$, and let $\{P_1, P_2, \dots, P_n\}$ be a partition of A such that $\mu(W_{A, P_1}) = 0$. Then:

$$(A, \{P_1, P_2, \dots, P_n\}, c) \succeq (A, \{P_1 \cup P_2, \dots, P_n\}, c') \iff c' \geq c,$$

with indifference when $c' = c$.

These two axioms capture invariance directly in the Expected Utility specialization of our framework:

PROPOSITION 2. *Suppose μ is regular, so that $(\Delta, (\Theta^\Delta, \Omega^\Delta, \mu), E^\Delta)$ is concordant. Then $\succsim$ satisfies B1 and B2 iff it satisfies Structural Invariance.*

Proof of Proposition 2. Necessity is straightforward. To prove sufficiency, we begin by proving a lemma.

LEMMA 3. *Let $\mathbf{W} \subseteq \Omega$ be μ -equivalent to $\mathbf{V} \subseteq \Omega$, and assume $\mu(W \cap W') = 0$ and $\mu(V \cap V') = 0$ for any distinct $W, W' \in \mathbf{W}$ and $V, V' \in \mathbf{V}$. Then there exists a bijection, q , between $\{W \in \mathbf{W} \mid \mu(W) > 0\}$ and $\{V \in \mathbf{V} \mid \mu(V) > 0\}$ such that $\mu(W) = \mu(W \cap q(W)) = \mu(q(W))$.*

Proof. Take some $W \in \mathbf{W}$ with $\mu(W) > 0$. By μ -equivalence, there exists a $V \in \mathbf{V}$ such that $\mu(W) = \mu(W \cap V) = \mu(V)$. To see that it is unique, let $V, V' \in \mathbf{V}$ both be such that the needed relation holds. Then we have $\mu(W) < 2\mu(W) = \mu(V \cap W) + \mu(V' \cap W) = \mu((V \cup V') \cap W) + \mu((V \cap V') \cap W) \leq \mu(W) + \mu(V \cap V')$. This means that $\mu(V \cap V') > 0$. By the condition in the statement of the Lemma, this requires $V = V'$. Reversing the roles of $\mathbf{W}$ and $\mathbf{V}$ yields the inverse map, so q is a bijection. ★

Let $(A, \{P_1, \dots, P_n\})$ and $(B, \{Q_1, \dots, Q_m\})$ be such that $\{W_{A, P_i}\}_{i \leq n}$ is μ -equivalent to $\{W_{B, Q_i}\}_{i \leq m}$. By Lemma 3, after relabeling the cells, we can assume that there are $1 \leq k \leq \min\{n, m\}$ positive-probability cells in each partition and that, for every $i \leq k$,

$$\mu(W_{A, P_i}) = \mu(W_{A, P_i} \cap W_{B, Q_i}) = \mu(W_{B, Q_i}).$$

³The reason there is a subset, rather than set equality, in the axiom is that it is possible that $r \in \alpha A + (1 - \alpha)A'$ is not a unique mixture of two elements. That is, $r = \alpha p + (1 - \alpha)q = \alpha p' + (1 - \alpha)q'$ for some $p, p' \in A$ and $q, q' \in A'$. For these elements, the cell of the partition in which they reside is not determined, but, it turns out not to matter.

Consider the problem $C = \frac{1}{2}A + \frac{1}{2}B$. Define $R_1 = \frac{1}{2}P_1 + \frac{1}{2}B$ and, recursively, for each $i = 2, \dots, n$,

$$R_i = \left(\frac{1}{2}P_i + \frac{1}{2}B \right) \setminus \bigcup_{j < i} R_j.$$

Then $\{R_1, \dots, R_n\}$ is a partition of C and $R_i \subseteq \frac{1}{2}P_i + \frac{1}{2}B$ for every $i \leq n$. Hence, by B1,

$$(A, \{P_1, \dots, P_n\}, c) \sim (C, \{R_1, \dots, R_n\}, c).$$

Now, define $R'_1 = \frac{1}{2}A + \frac{1}{2}Q_1$ and, recursively, for each $i = 2, \dots, m$,

$$R'_i = \left(\frac{1}{2}A + \frac{1}{2}Q_i \right) \setminus \bigcup_{j < i} R'_j.$$

Then $\{R'_1, \dots, R'_m\}$ is a partition of C and $R'_i \subseteq \frac{1}{2}A + \frac{1}{2}Q_i$ for every $i \leq m$. Hence, again by B1,

$$(B, \{Q_1, \dots, Q_m\}, c) \sim (C, \{R'_1, \dots, R'_m\}, c).$$

Let

$$N = \bigcup_{\substack{p, p' \in A \\ p \neq p'}} \left\{ \theta \in \Theta^\Delta : \sum_{\ell} \theta_{\ell} p_{\ell} = \sum_{\ell} \theta_{\ell} p'_{\ell} \right\} \cup \bigcup_{\substack{q, q' \in B \\ q \neq q'}} \left\{ \theta \in \Theta^\Delta : \sum_{\ell} \theta_{\ell} q_{\ell} = \sum_{\ell} \theta_{\ell} q'_{\ell} \right\}.$$

Since μ is regular and A and B are finite, $\mu(N) = 0$.

For every $i \leq n$ and $j \leq m$,

$$W_{C, R_i \cap R'_j} \setminus N \subseteq W_{A, P_i} \cap W_{B, Q_j}.$$

Indeed, if $\theta \notin N$ and $c_{\theta}(C) \in R_i \cap R'_j$, then $c_{\theta}(C)$ can be written both as $\frac{1}{2}p + \frac{1}{2}q$ for some $p \in P_i$ and $q \in B$, and as $\frac{1}{2}p' + \frac{1}{2}q'$ for some $p' \in A$ and $q' \in Q_j$. Since $c_{\theta}(C)$ maximizes utility over C , p and p' maximize utility over A , while q and q' maximize utility over B . Since $\theta \notin N$, these maximizers are unique. Hence $c_{\theta}(A) \in P_i$ and $c_{\theta}(B) \in Q_j$.

For each $i \leq k$, define

$$R''_i = R_i \cap R'_i.$$

If $j \neq i$, then

$$W_{A, P_i} \cap W_{B, Q_j} \subseteq W_{A, P_i} \setminus W_{B, Q_i},$$

and therefore

$$\mu(W_{A, P_i} \cap W_{B, Q_j}) = 0.$$

It follows that

$$\mu(W_{C, R_i \setminus R''_i}) = 0$$

for every $i \leq k$. Moreover, for every $i > k$, $\mu(W_{C, R_i}) = 0$, since $W_{C, R_i} \setminus N \subseteq W_{A, P_i}$. A symmetric argument gives $\mu(W_{C, R'_i \setminus R''_i}) = 0$ for every $i \leq k$, and $\mu(W_{C, R'_i}) = 0$ for every $i > k$.

Let

$$R^\dagger = C \setminus \bigcup_{i \leq k} R_i''.$$

By repeated applications of B2, in both directions, the zero-probability pieces can be split from the cells R_i and merged into $R^\dagger$. Hence,

$$(C, \{R_1, \dots, R_n\}, c) \sim (C, \{R_1'', \dots, R_k'', R^\dagger\}, c).$$

The same argument applied to $\{R'_1, \dots, R'_m\}$ gives

$$(C, \{R'_1, \dots, R'_m\}, c) \sim (C, \{R'_1'', \dots, R'_m'', R^\dagger\}, c).$$

Therefore,

$$(A, \{P_1, \dots, P_n\}, c) \sim (B, \{Q_1, \dots, Q_m\}, c).$$

It is easy to see that this implies $(A, \{P_1, \dots, P_n\}, c) \succeq (B, \{Q_1, \dots, Q_m\}, c')$ iff $c' \geq c$. ■

2 ADDITIONAL EXAMPLES

This section provides two additional examples that illustrate the role of concordance in different experimental environments. The first considers the textbook exercise of a modeler who wishes to infer a subject's preferences over a finite set. The second considers a modeler who wishes to learn a subject's preferences over a finite set but is constrained to ask the subject to choose from at most two binary menus, with the second menu potentially depending on the subject's first choice.

2.1 EXAMPLE: CHOICE CORRESPONDENCES

Example 1. Consider the textbook exercise of a modeler who wishes to infer a subject's preferences over some set X of cardinality n . Suppose her theory is that the subject is maximizing a complete and transitive preference and that she will observe what the subject's admissible choices out of each possible subset of X . Let $\mathcal{M}(X) = \{M_1, \dots, M_{2^n-1}\}$ be the set of all non-empty subsets of X and define the set of all possible choice patterns as

$$\mathbb{Z} = \{\times_{i=1}^{2^n-1} C_i \mid C_i \subseteq M_i \text{ for all } i\}.$$

The interpretation is that C_i is the set that contains all the alternatives the subject considers admissible when facing M_i .

Let Θ be the set of all weak orders over X (for $\theta \in \Theta$ write $x \succ_\theta y$ rather than $x\theta y$) and

$$c_\theta(\mathcal{M}(M_1), \dots, \mathcal{M}(M_{2^n-1})) = \times_{i=1}^{2^n-1} \{x \in M_i \mid x \succ_\theta y, \text{ for all } y \in M_i\}.$$

Let $\Omega = 2^\Theta$ and μ be an arbitrary probability measure on (Θ, Ω) .

PROPOSITION 3. Assume $\mathbb{E} = \{(\mathbb{Z}, \mathcal{P}) \mid \mathcal{P} \text{ is a partition of } \mathbb{Z}\}$, then the experimental paradigm $(\mathbb{Z}, (\Theta, \Omega, \mu), \mathbb{E})$ is concordant.

Proof of Proposition 3. First, observe that for any experiment $(\mathbb{Z}, \mathcal{P})$ and $P \in \mathcal{P}$, the set $\{\theta \mid c_\theta \in P\}$ is well defined and thus, (C1) holds. To show (C2) holds, take any partition $\mathcal{W} = \{W_1, \dots, W_k\}$ of Θ and define

$$P_{W_i} = \left\{ \prod_{j=1}^{2^n-1} \{x \in M_j : x \succeq_\theta y \text{ for all } y \in M_j\} \mid \theta \in W_i \right\} \text{ for all } i \leq k-1$$

$$P_{W_k} = \mathbb{Z} \setminus \bigcup_{i=1}^{k-1} P_{W_i}.$$

Since distinct weak orders induce distinct complete choice correspondences, $\mathcal{P} = (P_W)_{W \in \mathcal{W}}$ is a partition of $\mathbb{Z}$ that induces the partition $\mathcal{W}$ of Θ . Hence, $(\mathbb{Z}, \mathcal{P})$ is the experiment described by (C2). ■

Notice that if this environment were augmented so that the subject made a single selection out of each decision problem (i.e., the modeler observed a choice function, rather than a choice correspondence) then concordance does not hold. Indeed, it would be impossible to distinguish between subjects who strictly prefer x to y and who are indifferent but break ties in favor of x . In such a case, the algebra Ω would have to be coarsened to accommodate this lack of observability. □

2.2 EXAMPLE: ADAPTIVE EXPERIMENTS

To illustrate the constraints imposed by (C2), we apply our framework to adaptive experimental design. Broadly, the literature on adaptive design argues that when facing constraints such as time, money, or subject fatigue, a modeler may wish to use a dynamic procedure in which the questions posed to a subject depend on the subject's earlier answers (see Drake et al. (2022) and Chapman et al. (2024)).

Example 2. Consider a modeler interested in a subject's preferences over some finite set X of cardinality bigger than or equal to four, but who is constrained to ask the subject to choose from at most two binary menus. To maximize the information she can obtain, she uses the following adaptive procedure: she first offers a binary menu $\{a, b\}$; if the subject chooses a , then she offers $\{c_a, d_a\} \subseteq X$, otherwise she offers $\{c_b, d_b\} \subseteq X$ where $\{c_a, d_a\}$ need not be equal to $\{c_b, d_b\}$. Given the nature of the procedure, if the agent chooses a from $\{a, b\}$, then the modeler will know the choice out of $\{c_a, d_a\}$ but not the choice out of $\{c_b, d_b\}$, and analogously for a choice of b . Figure 1 illustrates the procedure.

We capture the choice patterns generated by a two-shot adaptive procedure by taking $\mathbb{Z} = X \times X$. Each adaptive experiment is then a pair $(A, \mathcal{P})$ where

$$A = \{(a, c_a), (a, d_a), (b, c_b), (b, d_b)\}$$

for some $a, c_a, d_a, b, c_b, d_b \in X$ and $\mathcal{P}$ is a partition of A .

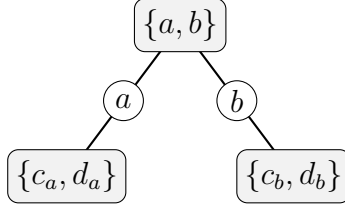


Figure 1: Adaptive Procedure

The modeler's theory about the subject's level of sophistication is crucial in this setting. To illustrate, consider first a modeler who assumes the subject is *naive*, choosing her preferred option in the first round without accounting for how it affects the second-stage choice. This can be captured by taking Θ to be a set of utility functions over X and defining

$$c_\theta(A) = (x, y) \quad \text{such that} \quad x \in \arg \max_{z \in \{a, b\}} \theta(z) \quad \text{and} \quad y \in \arg \max_{w \in \{c_x, d_x\}} \theta(w).$$

Next, consider a modeler who believes the subject is *sophisticated*. In this case, types are associated with the following choice behavior:

$$c_\theta(A) = (x, y) \in \arg \max_{(z, w) \in A} (\theta(z) + \theta(w)).$$

Finally, the modeler may wish to allow for both naive and sophisticated subjects, in which case Θ can encode not only the subject's utility function but also her level of sophistication.

In any case, the set of experiments $\mathbb{E}$ can only test a very limited set of hypotheses, and so in general (C2) will not hold. This is because, for any experiment in $\mathbb{E}$, there are at most four observable outcomes, thus precluding the identification of any finer partitions. For instance, the experiment predicated on $A = \{(a, c_a), (a, d_a), (b, c_b), (b, d_b)\}$, assuming all choice objects are distinct, cannot simultaneously determine a subject's preference between c_a and d_a and also between c_b and d_b . Notice that deeper trees of choices and highly constrained initial beliefs μ both offer a solution to this observability problem. $\square$

3 IDENTIFICATION VS CONCENTRATION

Our framework also allows for easy comparisons between prominent notions of informativeness. A belief-based criterion evaluates an identified set only through its ex-ante probability; Shannon entropy is the canonical example, $\tau(W) = -\log \mu(W)$, but nearly every widely used information-theoretic measure can define a belief-based index. By contrast, a posterior-concentration criterion values identification according to the induced posterior belief. The leading example is EC^2 , which, up to the common prior-squared term, values an identified set W according to $\tau(W) = \sum_{\theta \in \Theta} \mu(\theta \mid W)^2$.

To see how these criteria lead to different design choices, let $\Theta = \{\theta_1, \dots, \theta_7\}$ and suppose that $\mu(\theta_1) = \frac{2}{5}$, while $\mu(\theta_i) = \frac{1}{10}$ for each $i = 2, \dots, 7$. Consider two experiments,

A and B , inducing the following identifications

$$\mathcal{W}_A = \{\{\theta_1, \theta_2\}, \{\theta_3, \theta_4, \theta_5, \theta_6, \theta_7\}\} \quad \text{and} \quad \mathcal{W}_B = \{\{\theta_1\}, \{\theta_2, \theta_3, \theta_4, \theta_5, \theta_6, \theta_7\}\}.$$

Since A gives a perfectly balanced split of the prior, with induced probabilities $(\frac{1}{2}, \frac{1}{2})$ and B gives the less balanced split, $(\frac{2}{5}, \frac{3}{5})$, it is easy to see that the Shannon entropy criterion prefers A . A provides a safer reduction of uncertainty: whatever the subject's type, the modeler rules out half of the prior mass.

EC^2 ranks the same experiments differently. Under experiment A , the posterior following the first observation is $\mu(\theta_1|\{\theta_1, \theta_2\}) = \frac{4}{5}$ and $\mu(\theta_2|\{\theta_1, \theta_2\}) = \frac{1}{5}$, while the posterior following the second observation is uniform over five types. Under experiment B , the first observation identifies θ_1 exactly, while the second leaves a uniform posterior over the remaining six types. Ignoring the common prior-squared term, the EC^2 values are

$$\mathcal{U}^{EC}(A) = \frac{1}{2} \left[\left(\frac{4}{5}\right)^2 + \left(\frac{1}{5}\right)^2 \right] + \frac{1}{2} \left[5\left(\frac{1}{5}\right)^2 \right] = \frac{11}{25} < \frac{1}{2} = \frac{2}{5} \cdot 1 + \frac{3}{5} \left[6\left(\frac{1}{6}\right)^2 \right] = \mathcal{U}^{EC}(B)$$

Thus EC^2 prefers B , even though B is less balanced than A . The reason is that B has a positive probability of identifying the high-prior type θ_1 exactly, and EC^2 rewards the resulting posterior concentration.

This example also shows why the EC^2 criterion is not belief-based. Take the alternative experiment B' inducing the partition

$$\mathcal{W}_{B'} = \{\{\theta_1, \theta_2, \theta_3\}, \{\theta_4, \theta_5, \theta_6, \theta_7\}\}.$$

Just like B , B' yields an unbalanced split of $(\frac{3}{5}, \frac{2}{5})$. Thus, any belief-based representation would value B and B' equally; however, it is straightforward to see that $\mathcal{U}^{EC}(B) = \frac{1}{2}$ while $\mathcal{U}^{EC}(B') = \frac{2}{5}$.

These examples isolate a general tension. The distribution on Θ imparts two distinct features on an identified set: its size, captured by its prior probability $\mu(W)$, and its posterior composition, captured by $\mu(\cdot | W)$. Belief-based criteria such as Shannon entropy depend only on the first feature and therefore favor balanced identification. Posterior-concentration criteria such as EC^2 also depend on the second feature and may favor experiments that sharply isolate high-prior types. The EIV framework separates these two dimensions and therefore suggests hybrid objectives of the form

$$\tau(W) = G(\mu(W), \mu(\cdot | W)).$$

For example, for $\lambda \in [0, 1]$, one can take

$$\tau^\lambda(W) = \lambda[-\log \mu(W)] + (1 - \lambda) \left[\sum_{\theta \in \Theta} \mu(\theta|W)^2 - \sum_{\theta \in \Theta} \mu(\theta)^2 \right].$$

This nests Shannon entropy at $\lambda = 1$ and EC^2 at $\lambda = 0$, while intermediate values combine a concern for maximizing information gain and for posterior concentration.

4 DETAILS FOR SECTION 5.1 ON INTERTEMPORAL CHOICE

Here we show that

$$T(\mu_{A,z}) = - \left[\sum_{s=1}^{n+1} \pi_s(z) \frac{(\delta_{s-1})^2 + \delta_{s-1}\delta_s + (\delta_s)^2}{3} - \left(\sum_{s=1}^{n+1} \pi_s(z) \frac{\delta_{s-1} + \delta_s}{2} \right)^2 \right],$$

$$\bar{\sigma}_A(z) = \sum_{s=1}^{n+1} |I_s| \frac{B(\alpha + n - m(I_s, z), \beta + m(I_s, z))}{B(\alpha, \beta)}.$$

where

$$\pi_s(z) = \frac{|I_s| B(\alpha + n - m(I_s, z), \beta + m(I_s, z))}{\sum_{\ell=1}^{n+1} |I_\ell| B(\alpha + n - m(I_\ell, z), \beta + m(I_\ell, z))}.$$

Assume, as in the main text, that the cutoffs are ordered and write

$$0 = \delta_0 < \delta_1 < \dots < \delta_n < \delta_{n+1} = 1,$$

where $\delta_1, \dots, \delta_n$ are the cutoffs induced by the n binary questions. Let

$$I_s = [\delta_{s-1}, \delta_s), \quad s = 1, \dots, n+1.$$

On each interval I_s , the vector of discounted-utility prescriptions is constant. Thus, for each response pattern $z \in A$, $m(\delta, z) = m(\delta', z)$ for all $\delta, \delta' \in I_s$ for all s . We denote this common value by $m(I_s, z)$.

Fix a response pattern $z \in A$. Given type $\theta = (\delta, r)$, the likelihood of observing z is

$$\sigma_{\theta,A}(z) = r^{n-m(\delta,z)} (1-r)^{m(\delta,z)}.$$

Since δ is uniform on $(0, 1)$ and $r \sim \text{Beta}(\alpha, \beta)$ independently, the ex-ante probability of observing z is

$$\begin{aligned} \bar{\sigma}_A(z) &= \int_0^1 \int_0^1 r^{n-m(\delta,z)} (1-r)^{m(\delta,z)} \frac{r^{\alpha-1} (1-r)^{\beta-1}}{B(\alpha, \beta)} \, dr \, d\delta \\ &= \sum_{s=1}^{n+1} \int_{I_s} \int_0^1 r^{n-m(I_s,z)} (1-r)^{m(I_s,z)} \frac{r^{\alpha-1} (1-r)^{\beta-1}}{B(\alpha, \beta)} \, dr \, d\delta \\ &= \sum_{s=1}^{n+1} |I_s| \frac{B(\alpha + n - m(I_s, z), \beta + m(I_s, z))}{B(\alpha, \beta)}. \end{aligned}$$

Bayes' rule gives the posterior probability that δ lies in interval I_s after observing z :

$$\pi_s(z) = \mu_{A,z}(I_s \times [0, 1]) = \frac{|I_s| B(\alpha + n - m(I_s, z), \beta + m(I_s, z))}{\sum_{\ell=1}^{n+1} |I_\ell| B(\alpha + n - m(I_\ell, z), \beta + m(I_\ell, z))}.$$

Moreover, conditional on $\delta \in I_s$ and on observing z , the marginal posterior over δ is uniform on I_s . Indeed, the likelihood after integrating out r is constant in δ on each interval I_s ,

because $m(\delta, z) = m(I_s, z)$ for all $\delta \in I_s$. Hence the marginal posterior over δ is the finite mixture

$$\mu_{A,z,\delta} = \sum_{s=1}^{n+1} \pi_s(z) \text{Unif}(I_s).$$

It follows that

$$\mathbb{E}_{\mu_{A,z,\delta}}[\delta] = \sum_{s=1}^{n+1} \pi_s(z) \frac{\delta_{s-1} + \delta_s}{2},$$

and

$$\mathbb{E}_{\mu_{A,z,\delta}}[\delta^2] = \sum_{s=1}^{n+1} \pi_s(z) \frac{(\delta_{s-1})^2 + \delta_{s-1}\delta_s + (\delta_s)^2}{3}.$$

Therefore,

$$\text{Var}_{\mu_{A,z,\delta}}(\delta) = \sum_{s=1}^{n+1} \pi_s(z) \frac{(\delta_{s-1})^2 + \delta_{s-1}\delta_s + (\delta_s)^2}{3} - \left(\sum_{s=1}^{n+1} \pi_s(z) \frac{\delta_{s-1} + \delta_s}{2} \right)^2.$$

Since the posterior identification index is the negative posterior variance of the discount factor, we obtain

$$T(\mu_{A,z}) = - \left[\sum_{s=1}^{n+1} \pi_s(z) \frac{(\delta_{s-1})^2 + \delta_{s-1}\delta_s + (\delta_s)^2}{3} - \left(\sum_{s=1}^{n+1} \pi_s(z) \frac{\delta_{s-1} + \delta_s}{2} \right)^2 \right].$$

Finally, consider the deterministic benchmark $r = 1$. In this case, each response pattern identifies exactly one interval I_s . Conditional on observing the response pattern corresponding to I_s , the posterior over δ is uniform on I_s , and so

$$T(\mu_{A,z_s}) = -\text{Var}(\delta \mid \delta \in I_s) = -\frac{|I_s|^2}{12}.$$

Since $\mathbb{P}(\delta \in I_s) = |I_s|$, the expected posterior identification value is

$$\sum_{s=1}^{n+1} |I_s| T(\mu_{A,z_s}) = -\frac{1}{12} \sum_{s=1}^{n+1} |I_s|^3.$$

Equivalently, the expected posterior variance is

$$\frac{1}{12} \sum_{s=1}^{n+1} |I_s|^3.$$

This expression is minimized by equal-length intervals, yielding equally spaced cutoffs in the deterministic benchmark.

Finally, to see that the equal-spacing result need not survive noisy choice consider the case $n = 2$ and $r \sim \text{Beta}(2, 1)$. Substituting $\alpha = 2$ and $\beta = 1$ into the expressions above

and minimizing the expected posterior variance over $0 < \delta_1 < \delta_2 < 1$ yields

$$\delta_1^* = \frac{\sqrt{13} - 1}{6} \approx .434, \quad \delta_2^* = \frac{7 - \sqrt{13}}{6} \approx .566.$$

The resulting expected posterior variance is approximately 0.07257. By contrast, the equally spaced cutoffs $(1/3, 2/3)$ yield an expected posterior variance of approximately 0.07346. Thus, equal spacing is not optimal in this noisy-choice environment.

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